Course: Computational Linguistics 2

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Project titles for "Computational Semantics"

P1. Word Sense Disambiguation using Thesaurus/Dictionary/Knowledge-based Methods:

- Develop algorithms that leverage thesauri, dictionaries, or knowledge bases to determine the correct sense of ambiguous words in text.

P2. Word Sense Disambiguation Using Supervised Machine Learning Algorithms:

- Implement machine learning techniques to disambiguate word senses by training models on labeled data.

P3. Enhancing WordNet for Low-Resource Languages:

- Work on expanding and improving WordNet resources for languages with limited lexical data.

P4. Semantic Knowledge Representation and Reasoning: A First-Order Logic Approach:

- Develop a system that represents and reasons about semantic knowledge using first-order logic, exploring its applications in AI.

P5. Event-Based Meaning Representation for Temporal Understanding:

- Create models that represent and understand temporal aspects of meaning through event-based representations.

P6. Semantic Parsing using Compositional Semantics:

- Explore compositional semantics to develop parsers that convert natural language sentences into structured semantic representations.

P7. Exploring the Use of Graph-based Models in Semantic Parsing:

- Investigate the application of graph-based models to enhance semantic parsing, capturing rich semantic relationships.

P8. Cross-lingual Semantic Role Labeling: Adapting PropBank for Multiple Languages:

- Adapt PropBank annotations for semantic role labeling to work effectively across different languages.

P9. Comparative Analysis of LESK Algorithm with Other Word Sense Disambiguation Methods:

- Evaluate the performance of the LESK algorithm against other word sense disambiguation methods for comparison.

P10. Word Similarity/Relatedness using Lexical Resources

- Develop models to measure word similarity or relatedness using lexical resources such as WordNet, Synset, VerbNet, or FrameNet.

P11. Semantic Role Labeling-based NLP Application:

- Apply semantic role labeling techniques to build NLP applications like question-answering or machine translation systems.

P12. A Feature-based Algorithm for Semantic Role Labeling:

- Create algorithms for semantic role labeling that rely on feature engineering to enhance performance.

P13. Sentiment Analysis on Social Media Texts with Semantic Interpretation:

- Analyze sentiment in social media text while considering the semantic nuances and meanings behind expressions.

P14. Exploring Pretrained Language Models for Semantic Representations:

- Investigate the use of pretrained language models to capture semantic representations and apply them to various NLP tasks.

P15. Distributional Semantic Models for NLP tasks:

- Employ distributional semantic models in a range of NLP tasks, including text classification, named-entity recognition, metaphor detection, improving neural machine translation quality, multimodal solutions, and word sense disambiguation.

P16. A Feature-based Algorithm for Text Categorization:

- For a text categorization task of your choice, select a dataset and design features using naïve bayes and maximum entropy approaches to identify various categories associated with the dataset. Report precision, recall and accuracy figures along with a confusion matrix. Do a detailed linguistic analysis.

P17. Comparative Analysis of Word Embedding Techniques

- Evaluating performance of different word embedding methods (Word2Vec, GloVe, FastText) across multiple NLP tasks (at least 3) of your choice.

P18. Verifying Zipf's Law for Natural and Artificial Languages

For a natural language of your choice, verify Zipf's law for characters, words and morphemes. Subsequently generate an artificial language of the same length as the natural language using characters of the same language (space is also a character). Verify whether Zipf's law holds for the characters, words and morphemes of the artificial language. Write a report comparing the distribution of words for the natural and artificial language.

P19. Language Identification using Naive Bayes with Linguistic Features: A Comparative Study with Character N-gram Models

Choose data from language pair of your choice. Use a naive bayes classifier incorporating linguistically motivated features to identify these languages. Report precision, recall and accuracy figures along with a confusion matrix. Do a detailed linguistic analysis comparing the output of the naive bayes classifier with the simpler technique of character ngram frequencies that you implemented for the homework.

P20. Text Classification: Logistic Regression vs Naive Bayes

Take up a text classification dataset of your choice. Use a logistic classifier incorporating linguistically motivated features to identify various categories associated with this dataset. Report precision, recall and accuracy figures along with a confusion matrx. Do a detailed linguistic analysis comparing the output of the logistic regression classifier with the naive bayes classifier that you implemented for the homework.

P21. Sentiment Analysis with Naive Bayes and Lexicons

Take up a sentiment analysis dataset of your choice. Use a naive bayes classifier incorporating linguistically motivated features to identify various categories associated

with this dataset. Incorporate features based on a standard polarity lexicon of your choice for the task of sentiment analysis and report the results. Some popular resources:

1. General Inquirer (Stone et al., 1966)

2. LIWC (Pennebaker et al., 2007)

3. Opinion lexicon of Hu and Liu (2004)

4. MPQA Subjectivity Lexicon (Wilson et al., 2005).

Report precision, recall and accuracy figures along with a confusion matrix. Do a detailed linguistic analysis comparing the output of the classifier with the naive bayes classifier that you implemented for the homework.

P22. **Anaphora Resolution: Classifier vs Rule-Based**

Take up an anaphora resolution dataset of your choice. Implement a naive bayes or logistic regression classifier figure out the correct antecedent of each pronoun in the sentences. Do a detailed linguistic analysis comparing the output of your classifier with that of the rule-based system you implemented earlier.